

Dia-Shoe: A Smart Diabetic Shoe to Monitor and Prevent Diabetic Foot Ulcers

Nethmi Kularathne, Udara Wijayathilaka, Nethmi Kottawaththa, Shashika Hewakoralage, Samantha Thelijagoda
Faculty of Computing, Sri Lanka Institute of Information Technology

Malabe, Sri Lanka

nethmi2nethu@gmail.com, it16161702@my.sliit.lk, it16057920@my.sliit.lk, it16025004@my.sliit.lk, samantha.t@sliit.lk

Abstract—Diabetes is a continuous, lifetime disease that affects the body ability to use the energy found in food. Usually, the body splits down the sugars and carbohydrates you eat into a sugar called glucose. Glucose fuels the cells in the body. But the cells require insulin, a hormone, in the bloodstream to get in the glucose and utilize it for energy. Body doesn't make enough insulin, or it can't use the insulin it produces are the two main reasons to occur diabetes. Since the cells can't take in the glucose, it forms up in the blood. High levels of blood glucose can ultimately create nerve injury to the tissues in the extremities. The damage to nerves is a significant reason to occur foot ulcers. If a diabetic sufferer has foot ulcers, he/she need to manage it. Unless with the time they can drop their sensation in the foot which is known as neuropathy situation, which eventually appears in foot ulcers. Because of this, doctors advise using a diabetic shoe to prevent foot ulcers. These shoes decrease the peak plantar force exerted on the foot, but temperature, humidity and abnormal pressure are not managed, and this can be effective on the infection of wounds. As Diabetes is the main reason for the occurrence of these foot ulcers, maintaining blood glucose level is important. Diabetic patients can manage their blood glucose level by controlling daily calorie consumption and activities. For this self-maintenance of diabetes, patients should reduce their blood glucose level periodically and get notified of its divergence. The concept "Dia shoe" proposed is an innovative mobile base plug and play device which can be attached to any sort of diabetic shoe to manage the foot ulcers and prevent them from spoiling, while monitoring temperature, humidity, weight and step count of the patient through a mobile application.

Keywords—Linear Regression, Machine learning, Diabetes Mellitus

I. INTRODUCTION

Diabetes Mellitus or type-2 diabetes is one of the major non-communicable and fastest-growing public health problems in the world. From the world population, there are approximately 425 million adults (20-79 years) living with diabetes [1]. Apart from that, most of them suffer from diabetic foot ulcers, which are most significant and devastating complications of diabetes. It is estimated that about 5% of all patients with diabetes present with a history of foot ulceration, while the lifetime risk of diabetic patients developing this complication is 15% [2].

The root cause for occurring this kind of foot ulcers is the poorly controlled blood sugar level. This can damage the nerves and vessels that go to the feet, which increase the risk

of developing foot problems [2]. Diabetic neuropathy is a common factor in almost 90% of diabetic foot ulcers [2]. Motor neuropathy causes muscle weakness, while sensory neuropathy leads to loss of the protective sensation of pain, pressure, and heat [2]. Therefore, a remedy to control these ulcers is very important. There are specially designed diabetic shoes which reduce the peak plantar pressure and it contributes to the prevention of injuries associated with the diabetic foot. But the factors like temperature, humidity and vibration are also important factors that should be considered for the fast healing of wounds and keep the wounds uninfected.

Other than focusing on healing the wounds, overcoming the root cause for occurring these wounds is important. By properly maintenance of daily calorie intake and excising an individual can control his/her blood sugar level to a certain extent. Walking is recommended as an adjunct therapy to diet and medication in diabetic patients, with the aim of improving physical fitness, glycemic control and bodyweight reduction [3]. Therefore, tracking the number of steps walked per day is an important factor to be considered when focusing on the amount of calorie consumed and the amount of calorie burning. To help prevent foot problems, patients should wear comfortable, supportive shoes and take care of their feet before, during and after physical activity [4].

Although new treatments and technology have aided in controlling the disease in many individuals, the challenges of diabetes self-management are overwhelming for most. People with diabetes invest a lot of time and effort managing their condition including not only monitoring the level of sugar in their blood but organizing their medication and following a restrictive diet and exercise. Therefore, if the patient can track the deviation of his/her blood glucose level from the last checked sugar level, it will be helpful to the patient as he/she might not have to check the blood sugar regularly which ultimately encourage an individual to practice self-care behaviors.

The concept "Dia-Shoe" is a solution to overcome the above-mentioned issues. This proposed solution is a low-cost compact PCB device that can be plug and play into an existing specially designed diabetic shoe to monitor the status of the patient via a mobile device.

II. LITERATURE REVIEW

Contemporary researchers have come up with innovative solutions with various technologies to prevent foot complications. A group of researchers has prosperously designed a system to detect temperature, humidity and the plantar pressure across the foot. They have developed a smart shoe that monitors those three parameters across the foot and notifies the user about any significant change in those measurements. For self-monitoring purposes, the data is sent through a Bluetooth radio to an android app. Using a SurroSense Rx system they have identified 8 parts (hot spots) of the foot that can be supervised. For each hot spot, they have detected those three parameters. Researchers have used HIH6130 to measure temperature, humidity, and FlexiForce sensor to measure the plantar pressure. The Master device for the application is the embedded ATmega2560 on the Arduino board [15].

Some researchers have designed footwear for detecting the foot condition by measuring temperature, humidity, and foot pressure and if there are any abnormalities, this system will notify the concerned physician. The footwear has four sensors to detect the measurements. The detected data are given as inputs to the Arduino Uno microcontroller. Observed values are sent through the Raspberry Pi. The Raspberry Pi and the Arduino microcontroller is connected using a USB cable. These are fixed at the forefoot, hind foot, and the left-right of mid foot [3]. Placing the sensors and circuits in this way can be harmful to the foot ulcers of diabetic patients.

The area of physical activity plays a key role in controlling diabetes in diabetes patients and many research projects have been done related to this. According to the American Diabetes Association, regular activity is a key part of managing diabetes. It is said that when patients are active, their cells become more sensitive to insulin and can work more efficiently [8]. There are Android apps that have been developed to collect accelerometer data and calorie estimation in Android SDK [5]. Most of the applications use the accelerometer on the Android platform for identifying the physical activity a user is performing [6]. Researchers have used the accelerometer to measure the acceleration generated by the user's movements and used them in ACSM metabolic equations to calculate the number of calories burned. These applications also show the statistics of calories burned per day which can help people to control their weight anytime and anywhere [6]. Other than the accelerometer researchers have also used a pedometer, mobile phone GPS apps, sport GPS watches, smart watches and online maps to measure the walking distance.

To calculate calories burned or energy expenditure a human body needs to perform physical activities [5]. A number of calories burned per step depend on the height and weight of the patient. A typical 160-pound person of average height will burn about 40 calories per 1,000 steps and this is the equivalent of 0.04 calories per step [7]. To find the number of calories burned in each day, researchers have used ACSM metabolic equations. Maximal oxygen consumption equation was used for calculating the maximum oxygen consumption of a client's body (VO2Max) for a given exercise and the Caloric

expenditure equation was used for calculating the total number of calories burned from physical activity. Apart from that, the Basal Metabolic Rate (BMR) equation has also used for calculating the appropriate number of calories expended per day for each person [6].

There are Android calorie counting applications that help the user to keep track of their meals, exercise and weight. FatSecret (FatSecret, 2013) can be taken as an example of this. This application works by providing necessary information such as nutrition facts on foods and number of calories burned in each exercise mode [6]. The number of calories burned by the user's activities will be counted and stored in the application. Since the calorie counting process does not come from real practice, the results may be incorrect.

Most of the diabetic patients suffer from neuropathy. It is a type of nerve damage that can occur in diabetes patients resulting in pain and numbness in legs and feet. When this condition occurs, patients are not capable of feeling any injury or change in environmental parameters such as temperature, humidity, and pressure [15]. Dr John Dhanaseeley and V. Anusha Priya have shown that the foot infections of diabetic patients have different stages. If the infection is around 10-15%, then it is a mild infection which leads to removal of a finger in the feet. If the infection is about nearly 50%, then it is a serious infection which may cause the removal of the leg [3]. Measuring the temperature, pressure, and humidity of the foot can help mitigate these risks of foot ulceration.

Humidity is the amount of water vapor available in the air [15]. The relative humidity is expressed as a percentage of the amount needed for saturation at the same temperature. The relative humidity is directly related to temperature. Thus, it affects how the human body feels with respect to the environment [15]. When the relative humidity is high, we feel much hotter than the actual temperature. Diabetes hampers the ability of the body to cope with high heat and humidity. There are factors in temperature and humidity levels that are affected by the diabetic foot ulcers, such as the behavior of human skin, activities and climate.

In 2010, a group of researchers have studied the relationship between diabetic neuropathy patients and their foot temperature [16], which has shown that the neuropathy patients had the highest foot temperature (32^o C– 35^o C) compared to the foot temperature of non-neuropathy diabetes patients (27^o C– 30^o C) and, diabetic neuropathy patients had higher mean foot temperature (MFT) compared to the non-neuropathic patients [16].

There are many major factors that can be affected by high and low blood glucose level. BMI, age, race, genetic factors, smoking, stress level (Cortisol hormones), measured fasting blood glucose or Hemoglobin A1c (HbA1c) level, daily exercise, the amount of consumed dietary carbohydrate are major factors that can cause diabetes [12].

Being overweight increase the risk of occurring type 2 diabetes and it can also increase the risk of high blood pressure [8]. According to researches, people with higher BMI are at higher risk of having diabetes mellitus. Diabetic can occur if you are inactive, overweight or obese [9]. Sometimes,

extra weight can cause insulin resistance and it is common in people with type 2 diabetes [10]. Many factors affect type 2 diabetes. Some of the factors like body weight and how much the patient exercise can be changed while factors like genetic factors are unchangeable [11].

Exercise is a key factor in managing blood sugar level in diabetic patients and it has an ability to reduce the stress level of the patients. Physical activity can maintain the blood sugar level to a lower level. If the patient exercises regularly, the cells may be more sensitive to insulin. Then the patient can keep blood sugar levels within the normal range [14]. The less active the patient is, the greater the risk of type 2 diabetes. Physical activity helps you control your weight, uses up glucose as energy and makes your cells more sensitive to insulin [12].

Nicotine can make your blood sugar level go up or down. It has found that diabetes and smoke need larger doses of insulin to control their blood glucose. Nicotine changes chemical processes in your cells, and they respond to insulin [13]. Because of this smoking raise blood sugar and it can lead to higher blood sugar levels of patients [14].

There are many related applications which are used throughout the world to assist diabetic patients. But these systems have unresolved problems with it. When correlating these applications with our application, Dia-Shoe is invented specifically to use with the compact device that should be wear with the shoe. Most of the functions of the application will depend on the readings received from the device.

III. METHODOLOGY

The aim of our research is to prevent diabetic foot ulcers of type 2 diabetic patients. This developed solution is low power consumption, a low-cost smart diabetic shoe that monitors step count, the weight of the patient, temperature and humidity of the shoe.

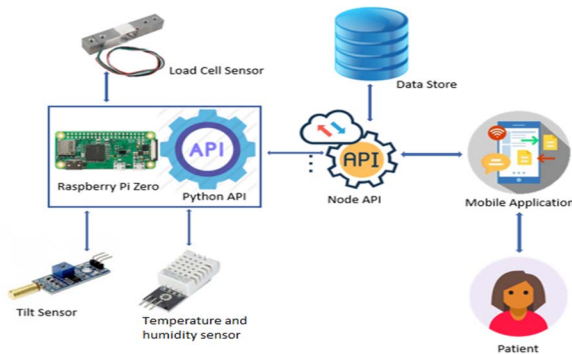


Fig.1. System Architecture Diagram

Our team has developed a multivariate extendable sensor platform that can gather temperature, humidity, step count and the weight sensor details in real-time for later analysis. A low-cost Raspberry pi zero w single-board computer is used here. The Raspberry Pi zero w performs data acquisition, control, formatting, and recording. This single-board computer was chosen due to the high connectivity, availability and low cost. The language used to develop the system inside the Raspberry

Pi Zero is the python language and the "sci-kit-learn library" is used for machine learning implementations.

Linear regression algorithm is used to minimize the sensor errors and increase the accuracy of the three sensors. Simple linear regression is a type of regression analysis algorithm where the number of independent variables is one and there is a linear relationship between the independent (x) and dependent (y) variable.

The sensor readings from the three sensors are taken and the actual readings are taken from a laboratory thermometer, hygrometer, accelerometer and a weighing scale. Based on the actual readings and the sensor readings, a line is plotted which models the points to the best. This line is modeled based on the linear equation.

Using Cost function, the best possible value is found which would provide the best fit line for the data points. Cost function helps to find out the best possible values which provides the best fit line for the data points to minimize the problem. Likewise, errors have been minimized between the sensor values and the actual values.

The root cause for occurring diabetic foot ulcers is the poorly controlled blood sugar level. The factors like temperature and humidity are also key factors that should be considered for the fast healing of wounds and keep the wounds uninfected. For the above problems, our team has developed a solution and to come up with that solution, various kind of predictions have done by collecting the data sets. Therefore, in the beginning, datasets have been collected using calibrated sensors.

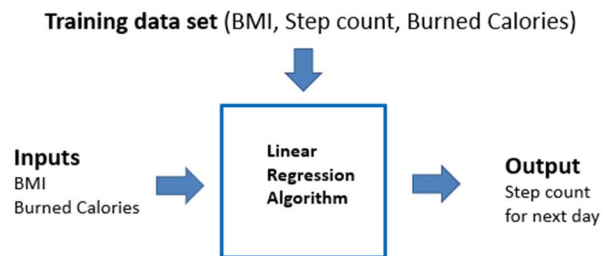


Fig.2. Mechanism of step count prediction

First, the target step count for the next day is predicted based on daily calorie consumption. The mechanism of step count prediction is given in the above Fig.2. Our team collected a dataset by participating fifty people who are aged between 40 and 50 (both men and women). The "Tilt switch" sensor was used to monitor the number of steps walked by a person daily. The sex, height and weight of the patient were also required for the predictions. The sex and height were taken at the time of sign up for the application. The weight of the patient is measured with the use of a calibrated load cell sensor. Since the accuracy depends on the patterns presented, behavior and the size of the training dataset, the Linear Regression Machine Learning algorithm is used for this prediction.

To predict the deviation of Blood Glucose Level from existing blood glucose level, a data set is needed, and it was created by considering the same patients mentioned above. The age, current blood glucose level, genetic factors, smoking, HbA1c report details which are given by a doctor and the amount of consumed dietary data are collected from these people. The mechanism of blood glucose level deviation prediction is given in Fig.3.

At the time of signup to the application, the patients are requested to enter all this information. When the patient adds his/her height, the BMI value is calculated with the use of weight taken from the "load cell" sensor. The data related to smoking and genetic factors are in binary format. The current blood glucose level is taken using a glucometer or by doing a blood test.

To analyze the temperature and humidity inside the shoe, a dataset is created. For this, the temperature and humidity readings are taken from diabetic patients who are suffering from foot ulcers and wearing diabetic shoes. The mechanism of predicting the time to reach the critical temperature inside the shoe is given in Fig.4. A DHT22 sensor is used to gather temperature and humidity values and this sensor is used to read both temperature and humidity readings. during a specific time.

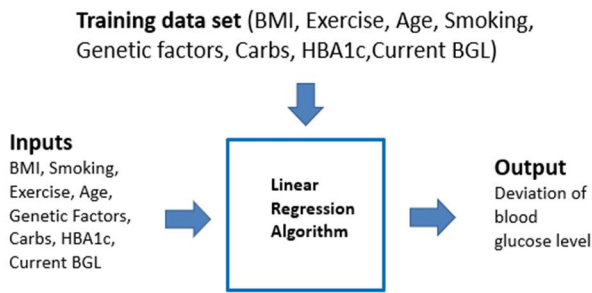


Fig.3. Mechanism of Blood glucose level deviation prediction

The data processed using these algorithms are transmitted to an android application via Wi-Fi. The application was developed using React Native which has a node backend. The processed data is transmitted to the node API via the python

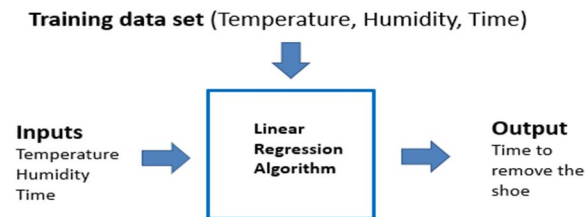


Fig.4. Mechanism of predicting the time to reach the critical temperature

API in the form of JSON objects.

IV. RESULTS

We implement our Diabetes Shoe design in five stages of development where each successful stage leads to the start of the next stage. The five stages of development are described as following. Calibrating the three sensors (DHT22, load cell and tilt sensor), setup the Raspberry Pi Zero with a Raspbian OS using a 16GB SD, connect the sensors to the Raspberry Pi, building the prototype circuit, and manufacturing and assembling the final PCB.

A. Overall Diabetes Shoe Results

The readings for the load cell was obtained using different standard weights while the tilt sensor was calibrated using the adafruit sensor library. The room temperature and humidity were measured using a thermometer and a hygrometer and using the DHT22 sensor. Finally, the collected data set was tracked using the linear regression algorithm to obtain the respective sensor calibration accuracies. The resulted mean squared error (MSE), R2 score, and variance score are given below.

TABLE I: Accuracies of sensor calibration for load cell, tilt sensor and DHT22 sensor using linear regression algorithm

	Result				
	MSE	R2 score	Variance Score	MSE Significant	R2 score and variance score significant
Load cell	0.034	0.8792	0.9234	0.05	0.80
Tilt sensor	0.028	0.9267	0.9489	0.05	0.80
DHT22 (Humidity)	0.013	0.9892	0.9123	0.05	0.80
DHT22 (Temp)	0.019	0.8923	0.8967	0.05	0.80

The Raspberry Pi zero was configured with a Raspbian OS using a 16GB SD and the three sensors were connected to the Raspberry Pi to build the prototype circuit. Fig. 5 shows the prototype model build.



Fig.5. Prototype model of the device

We generate the Gerber files required for PCB printing. Fig. 6 shows the design of the PCB. We manufactured a diabetic shoe with a core to embed the circuit. Finally, we connect the sensor to the final PCB and inserted the device into the diabetic shoe. The Fig. 7 shows our final Diabetes Shoe.

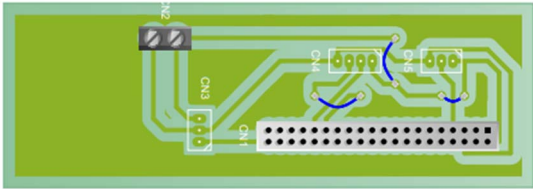


Fig.6. Printed Circuit Board (PCB)



Fig.7. Final diabetes shoe with the device

B. Overall Mobile Application Results

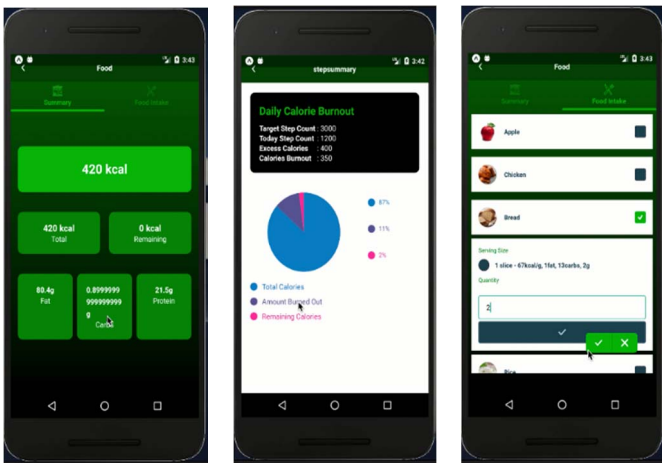


Fig.8. Interfaces of ‘food’ and ‘step count’ features.

First, the user logs in to the application and enter his or her personal information and medical status. After confirming these details, the user can select a feature from a range of features from the dash board. For instance, if the user selects the feature ‘food’, the user can add the his/her daily food intake and view the total calorie consumption upon the recommended level. As the user navigates to the feature ‘step count’, the user is presented with a pie chart indicating the latest calorie burn

out and remaining amount. This feature also facilitates the user by predicting the step count that the user should walk on next day. The above fig.8 shows the main interfaces of the ‘food’ and ‘step count’ features.

If the user selects the feature ‘BGL deviation’ the user can view his/her BMI level and moreover, the application offers the user the facility to predict his/her blood glucose level deviation respective to the last updated actual blood glucose level. Furthermore, this application is designed specifically to monitor the temperature and humidity inside the shoe. If the user selects the feature ‘Temp and humidity’, the user can view the remaining time to remove the shoe depending on the temperature and humidity levels inside the shoe. This feature helps to maintain a favorable environment to wounds inside the shoes and prevent infections. The fig.9 shows the main interfaces of the ‘BGL deviation’ and ‘Temp and humidity’ features.

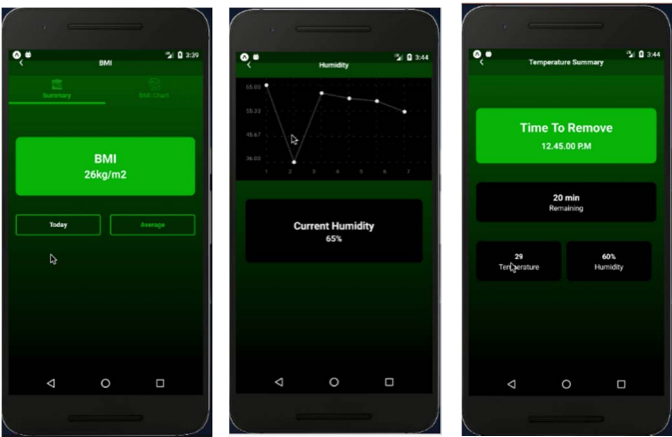


Fig.9. Interfaces of ‘BGL deviation’ and ‘Temp and humidity’ features.

For the predictions of the step count, blood glucose level deviation and the time to remove the shoe the data gathered from sensors and trained using linear regression algorithms. The Fig.2, Fig.3 and Fig.4 show how the prediction was done. This training was done inside the Raspberry Pi Zero which has a high processing speed and a memory. The resulted output was transmitted to the node API of the mobile via a python API using JSON objects. The below tables show the accuracy and mean squared error of the trained data set obtained for each of the three sensors.

TABLE II: Accuracies of step count prediction using Linear regression algorithm.

	Result				
	MSE	R2 score	Variance Score	MSE Significant	R2 score and variance score significant
Step count	0.0282	0.7184	0.8186	0.05	0.80
Burned Calories	0.0082	0.7802	0.8803	0.05	0.80

TABLE III: Accuracies of time prediction to remove the shoe depending on temperature and humidity by using linear regression algorithm.

	Results				
	<i>MSE</i>	<i>R2 score</i>	<i>Variance Score</i>	<i>MSE Significant</i>	<i>R2 score and variance score significant</i>
Temperature	0.0140	0.9832	0.9842	0.05	0.80
Humidity	0.0243	0.8975	0.8977	0.05	0.80

TABLE IV: Accuracies of blood glucose level deviation prediction using linear regression algorithm.

	Results				
	<i>MSE</i>	<i>R2 score</i>	<i>Variance Score</i>	<i>MSE Significant</i>	<i>R2 score and variance score significant</i>
Blood glucose deviation	0.0150	0.7834	0.7346	0.05	0.80

MSE stands for mean squared error. To get a better output MSE must be a lower value. Also, R2 score and variance score must be greater than 0.90. According to the values in the above tables, the predicted outputs are more accurate.

V. CONCLUSIONS & FUTURE DIMENSIONS

We have developed a prototype smart shoe to monitor and prevent diabetic foot ulcers for type 2 diabetic patients. This will be beneficial in all the diabetic population who suffers from diabetic foot ulcers. The plug and play device developed by us can be used by the users along with the android application installed to maintain their diabetes.

Recommendations to those who are willing to develop this system further are as follows.

- Setting up a navigation map to identify the current location of the patient.
- Increase the accuracy of the blood glucose level deviation prediction.
- Sending a SMS to a patient's relative when the blood glucose level of the patient exceeds the normal level.
- As the research was limited to a specific time period our team was able to focus only a limited number of components.

In the future, we are willing to do more researches and develop more methods that will be included in the hardware device as well as to the android application.

REFERENCES

- [1] Leonor Guariguata, David Whiting, Clara Weil, Nigel Unwin, (2011).The International Diabetes Federation diabetes atlas methodology for estimating global and national prevalence of diabetes in adults.ScienceDirect 3,322-332.
- [2] R. L. Harries and K. G. Harding, "Management of Diabetic Foot Ulcers," *Current Geriatrics Reports*, vol. 4, no. 3. pp. 265–276, 2015.
- [3] V. A. P. D. J. Dhanaseel.A2, 1, and I. Department of ECE, IFET college of Engineering and Technology, Villupuram, "HEALTH CARE FOOT WEAR FOR MONITORING THE DIABETIC PATIENTS," 2018.
- [4] K. R.V., V. D. R.W.M., H. K., and P. P., "Walking performance in people with diabetic neuropathy: Benefits and threats," *Diabetologia*, vol. 49, no. 8. pp. 1747–1754, 2006.
- [5] S. Kashi, C. Reddy, and P. Reddy, "Fit Assist Step count and Calorie Estimator using Accelerometer," vol. 3, no. 9, pp. 7912–7917, 2014.
- [6] S. Sukaphat and C. S. Program, "an Applying of Accelerometer in Android," no. May, pp. 294–304, 2013.
- [7] W. Bumgardner, "Pedometer steps to calories converter," *VeryWell*. 2016.
- [8] American Diabetes Association, "Overweight: American Diabetes Association®." 2016.
- [9] H. E. Bays, R. H. Chapman, and S. Grandy, "The relationship of body mass index to diabetes mellitus, hypertension and dyslipidaemia: Comparison of data from two national surveys," *International Journal of Clinical Practice*, vol. 61, no. 5. pp. 737–747, 2007.
- [10] The National Institute of Diabetes and Digestive and Kidney Diseases.(2016)Symptoms & Causes of Diabetes [Online]. Available:https://www.niddk.nih.gov/health-information/diabetes/overview/symptoms-causes.
- [11] "Age, Race, Gender & Family History_ American Diabetes Association®." .
- [12] Mayo Clinic Staff, "Multiple myeloma - Symptoms and causes - Mayo Clinic," 2017.
- [13] Minesh Katri, "Nicotine's Effect on Blood Sugar Levels," *American Diabetes Association*. 2017.
- [14] Peggy Pletcher.(2016)Smoking and Diabetes: 4 Smoking-Related Problems [Online].Available:https://www.healthline.com/health/type-2-diabetes/smoking-diabetes.
- [15] B. P. Olawale David Jegede, Ken Ferens, Bruce Griffith, "(PDF) A Smart Shoe to Prevent and Manage Diabetic Foot Diseases," *Int'l Conf. Health Informatics and Medical Systems*. pp. 47–53, 2015.
- [16] Subramnaia Bagavathiappan, John Philip, "Correlation between plantar foot temperature and diabetic neuropathy: a case study by using an infrared thermal imaging technique," *Journal of diabetes science and technology*, vol. 4, no. 6. pp. 1386–1392, 2010.